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Formalizing the Real Numbers in Homotopy Type Theory with Cubical Agda

- 9,000 lines of code
- 50+ definitions, lemmas, and theorems
- Pull request to agda/cubical

```
2+2=4 : 2 + 2 ≡ 4
2+2=4 = refl
```

Cubical Agda:

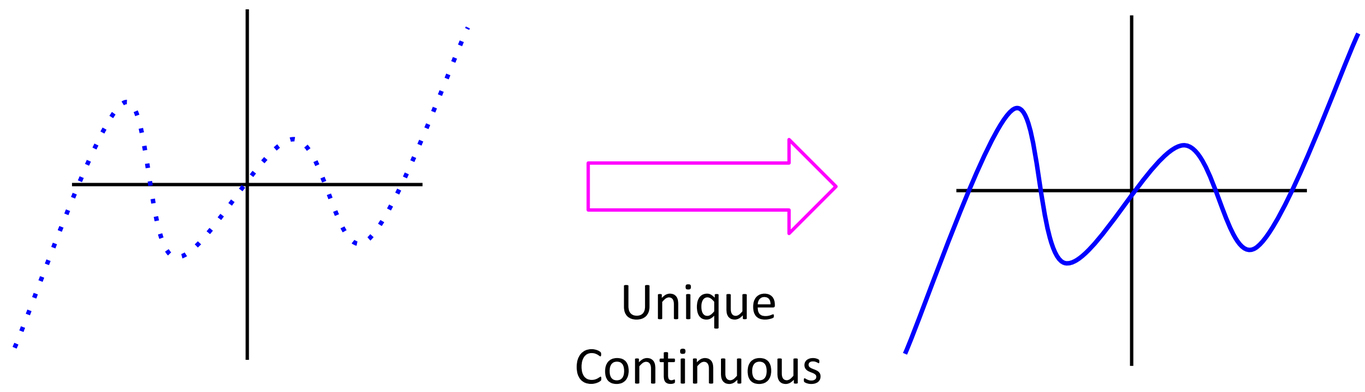
```
"Yep, I ran the left-hand side
and it reduces to 4, lgtm"
```

Does not compute

```
example : (2 : R) + (2 : R) =
          (4 : R) :=
    by rfl
       ...
```

Lean:

```
Tactic `rfl` failed:
  2 + 2
is not definitionally equal to
  4
```



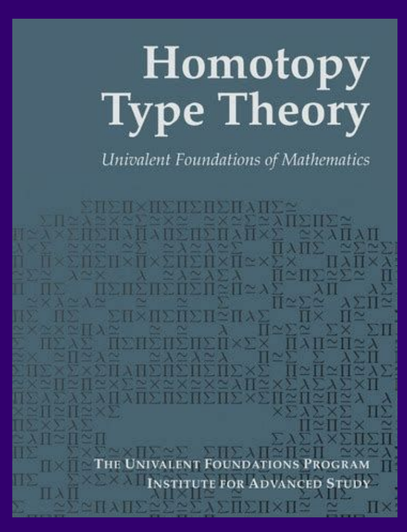

```
_+_ : R → R → R
_+_ = liftNonexpanding₂ Q._+_
      +nonexpandingQ₂

max : R → R → R
max = liftNonexpanding₂ Q.max
      maxNonexpandingQ₂

_≤_ : R → R → Type ℓ-zero
_≤_ x y = max x y ≡ y
```

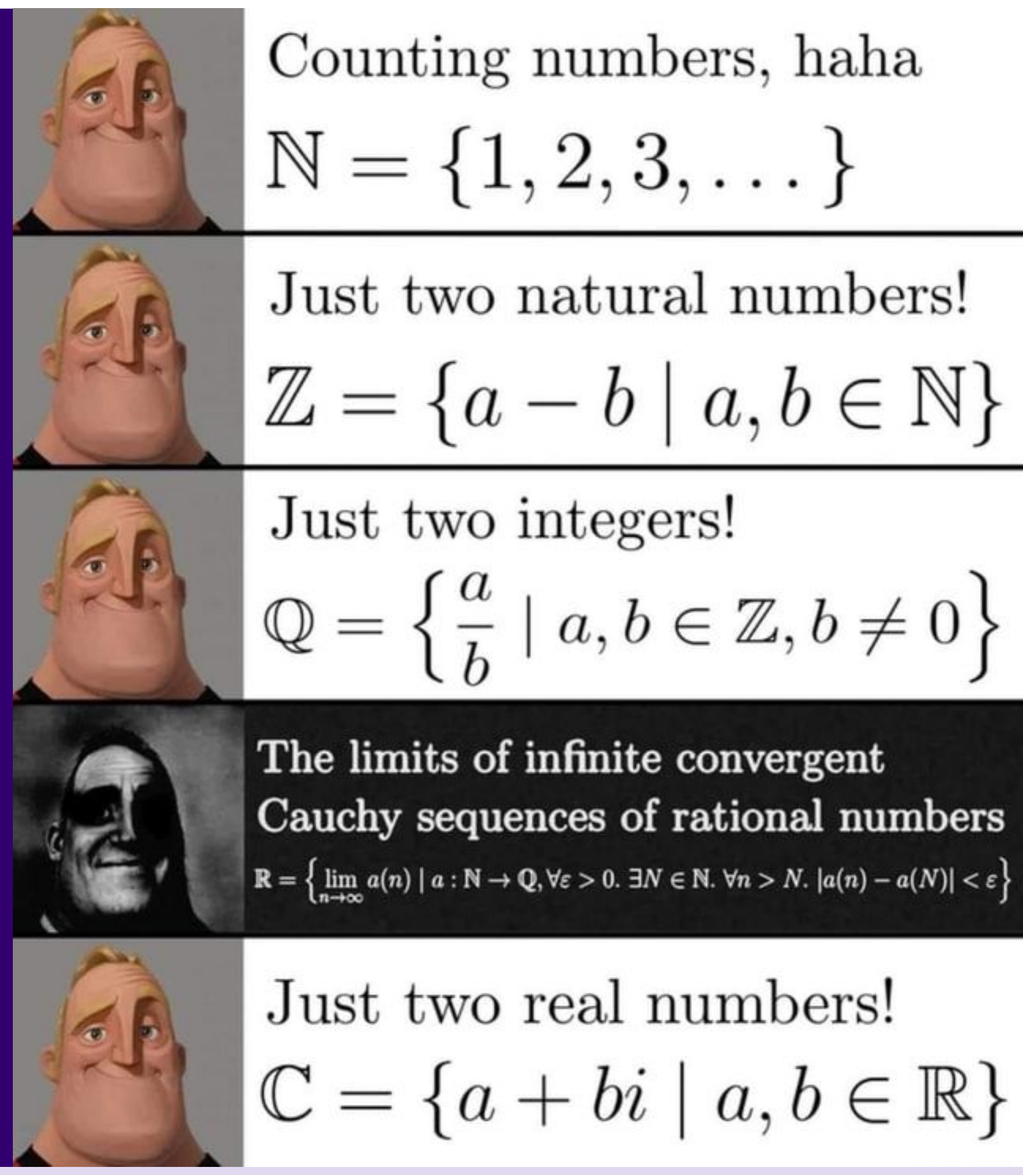
Why a new construction of the reals?

Bishop	Regular Cauchy	Dedekind
❌ "Setoid hell"	❌ ax. countable choice	❌ Universe levels



Solution: Use a higher inductive type:
+ Equality captured directly by identity types
+ Avoid countable choice
+ Cauchy representation

"Any foundation of mathematics worthy of its name must eventually address the construction of real numbers" (HoTT 2013)"




```
data R : Type ℓ-zero

data Close :
  (ε : Q) → (0 < ε) → R → R → Type ℓ-zero

data R where
  rational : Q → R
  limit : (x : (ε : Q) → 0 < ε → R) →
          CauchyApproximation x →
          R
  path : (x y : R) →
         ((ε : Q) (p : 0 < ε) →
          x ~[ ε , p ] y) →
         x ≡ y
```

```
CauchyApproximation :
  ((ε : Q) → 0 < ε → R) → Type ℓ-zero
CauchyApproximation x =
  ((δ ε : Q) (p : 0 < δ) (q : 0 < ε) →
   x δ p ~[ δ + ε , 0<+' p q ] x ε q)
```

Closeness is an inductively defined relation (code too big to fit):

- for any q, r, ϵ, if $-\epsilon < q - r < \epsilon$, then $\mathsf{rat}(q) \sim_\epsilon \mathsf{rat}(r)$,
- for any q, y, ϵ, δ, if $\mathsf{rat}(q) \sim_{\epsilon-\delta} y_\delta$, then $\mathsf{rat}(q) \sim_\epsilon \lim(y)$,
- for any x, r, ϵ, δ, if $x_\delta \sim_{\epsilon-\delta} \mathsf{rat}(r)$, then $\lim(x) \sim_\epsilon \mathsf{rat}(r)$,
- for any $x, y, \epsilon, \delta, \eta$, if $x_\delta \sim_{\epsilon-\delta-\eta} y_\eta$, then $\lim(x) \sim_\epsilon \lim(y)$,
- for any u, v, ϵ, if $\xi, \zeta : u \sim_\epsilon v$, then $\xi = \zeta$ (propositional truncation).